Business Case

Demand Model Optimization at RetailCo

1. Introduction: RetailCo's Story

RetailCo is a fast-moving consumer goods company founded in 2015 in Latin America, specializing in the commercialization of high-turnover products. In less than a decade, the company has built a regional presence with a distribution network spanning three distinct sales channels.

Sales Channels

Channel	Market Share	Description
Supermarkets	45% of volume	Large retail chains with weekly supply contracts and structured planning.
Traditional	35% of volume	Neighborhood stores and wholesale distributors with irregular orders and volatile demand.
E-commerce	20% of volume	Own platform and marketplaces with high seasonal variability and algorithmic price sensitivity.

Product Catalog

The company markets five (5) products distributed across two main categories, each with different shelf lives that represent an inventory management challenge:

Category	Product	Presentation	Shelf Life
Beverages	Natural Juice	1 L	45 days
Beverages	Flavored Water	500 ml	60 days
Beverages	Energy Drink	350 ml	90 days
Snacks	Whole Grain Crackers	200 g	30 days
Snacks	Cereal Bar	50 g	40 days

Current Forecasting Model

RetailCo operates with a forecasting model based on a Machine Learning algorithm called LGBM (Light Gradient Boosting Model). This model predicts Sell-In sales in volume (Tonnes) using additional variables such as: Price per kg, Numeric Distribution, Weighted Distribution, Advertising Investment (USD), Promotion Investment (USD), Promotion Type, Promotional Dates, and Holiday Calendar. Once the model generates a total forecast, it applies the SHAP Values methodology to obtain an explanation of how each input driver contributes to the total figure — this is referred to as forecast decomposition.

The modeling process uses a single set of general parameters, but during training, a segmentation (Slicing) is applied by Channel. This means a separate model is trained for each channel, all sharing a common set of parameters, variables, and feature engineering rules.

2. Problem Statement

Over the last six (6) months, RetailCo has faced two critical problems directly attributed to the performance of its demand model:

Problem 1: Write-Off Losses due to Product Expiration

The model shows an average accuracy of 62% (measured as 1 − MAPE) at the weekly SKU-Channel level, well below the industry benchmark (75–85%). This low precision has generated:

1. Recurring overstock in the Traditional channel for the Snacks category, where Write-Off reached 8.2% of production volume in the last quarter (target: < 2%).
2. Stock-outs in E-commerce for Beverages during demand peaks not captured by the model.
3. Accumulated losses of $1.2M USD over the last 6 months due to expired product in warehouses and at point of sale.

Indicator	Current Value	Target / Benchmark
Model Accuracy (1 − MAPE)	62%	75–85%
Write-Off (Snacks / Traditional Channel)	8.2% of volume	< 2%
Expiration Losses (6 months)	$1.2M USD	—

Problem 2: Poor Interpretability of the Forecast Decomposition

The commercial team uses the model's decomposition to make trade marketing investment decisions. However, structural inconsistencies have been detected:

1. The "Price" variable shows a positive contribution to volume in the Supermarkets channel for Natural Juice 1 L, which contradicts economic logic (price-demand elasticity should be negative).
2. The "Promotions" variable captures a disproportionate effect (40% of explained variation) in E-commerce, when the real primary driver is Advertising Investment, which should not exceed 20% of the base.
3. The commercial team increased promotional investment in E-commerce by 35% three months ago without achieving incremental returns.
4. Three months ago, the price of Natural Juice in Supermarkets was reduced expecting a small effect on demand, which turned out to be null.

Variable	Observed Anomaly	Commercial Consequence	Affected Channel / SKU
Price	Positive sign (incorrect elasticity)	Price reduction with no impact on sales	Supermarkets / Natural Juice 1 L
Promotions	40% variation over-attribution	+35% investment without incremental return	E-commerce / Beverages

3. The Demand Planning Meeting

In two weeks, the monthly Demand Planning meeting will take place, attended by leaders from different areas (Marketing, Finance, Sales, Operations) to address the problems identified above. To this end, they have invited you — a data scientist at the company — to help answer the following questions:

1. What is the current status of the forecasting and decomposition system?
2. What can be done to improve the accuracy of the systems?
3. How do you ensure that the proposed solutions will work? And to what extent will they resolve the issues?
4. What would be the next steps to continue with your plan, quick wins, and risks to consider?

4. Available Information

To solve this Business Case, the candidate will have access to the following information, which will be detailed in a complementary Excel file accompanying this document:

1. Weekly Sell-In sales history by Product and Channel for the last 3 years (156 weeks).
2. History of external variables by Product and Channel for the last 3 years (156 weeks), plus projected data for the next 18 months (78 weeks).
3. Current model parameters.
4. Definition of the feature engineering rules used.
5. Feature importance information for each trained model (recall: one model per channel).
6. Results of the current decomposition (contribution of each variable per SKU-Channel).

5. Presentation Day

For the day of the meeting, please prepare a presentation (1 hour) covering the following points:

- A quick introduction about yourself (5 min).
- An executive summary for your presentation (10 min) — it should include the problem statement.
- The answer to the 4 questions (30 min in total).

Keep in mind the following:

- Your audience is not a technical audience.
- There is no single "correct" answer — we evaluate your reasoning process.
- If anything is not clear or incomplete in the material, feel free to make any assumption.
- Document must be elaborated in English.
- Presentation can be in Spanish or English.

Dataset Tables

Attached to this document you will receive an Excel file called "Business Case Data Set". That file contains 7 tables:

- **Table 0:** Data Dictionary with the information of each table and column.
- **Table 1:** Information on external variables (past and future).
- **Table 2:** Sell-In information for the past.
- **Table 3:** Model Parameters.
- **Table 4:** Feature Engineering Rules.
- **Table 5:** Feature Importance.
- **Table 6:** SHAP Values.
- **Table 7:** Forecast Results considering current external variables and model setup.